

$$z_A = Z \left(\begin{array}{c} A \\ \text{cup} \\ A \end{array} \right) = \left(A \xrightarrow{1 \otimes \text{coev}} A \otimes B \otimes A \xrightarrow{\text{ev} \otimes 1} A \right) : A \rightarrow A$$

$$z_B = Z \left(\begin{array}{c} B \\ \text{cup} \\ B \end{array} \right) = \left(B \xrightarrow{\text{coev} \otimes 1} B \otimes A \otimes B \xrightarrow{1 \otimes \text{ev}} B \right) : B \rightarrow B$$